Anirudh Buvanesh

Ph.D. Student, Université de Montréal and Mila — reinforcement learning for reasoning in large language models
Previously: large-scale retrieval at Microsoft Research India, deployed in Bing Search and Sponsored Search

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Education

Université de Montréal *Aug 2024 – Aug 2029*
Ph.D. in Computer Science **GPA: 4.3 / 4.3**
Advisors: Prof. Aaron Courville and Prof. Laurent Charlin

Birla Institute of Technology and Science (BITS), Pilani *2016 – 2021*
M.Sc. Mathematics and B.E. Computer Science **GPA: 9.36 / 10 (Distinction)**
Undergraduate thesis: *Deep Learning Based Architecture for Agriculture Vision* **Department rank: 3 / 70**

Publications

On the Necessity of World Knowledge for Mitigating Missing Labels in Extreme Classification
Jatin Prakash*, **Anirudh Buvanesh***, Bishal Santra, Deepak Saini, Sachin Yadav, Jian Jiao, Yashoteja Prabhu, Amit Sharma, Manik Varma
*ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD**), 2025*
Uses product information to make recommendations despite missing user feedback; **deployed in Microsoft Sponsored Search.**

Extreme Meta-Classification for Large-Scale Zero-Shot Retrieval
Sachin Yadav*, Deepak Saini*, **Anirudh Buvanesh***, Bhawna Paliwal, Kunal Dahiya, Siddarth Asokan, Yashoteja Prabhu, Jian Jiao, Manik Varma
*ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD**), 2024*
Recommends newly added products with no interaction history; **deployed in Bing** and led to a granted patent.

Enhancing Tail Performance in Extreme Classifiers by Label Variance Reduction
Anirudh Buvanesh*, Rahul Chand*, Jatin Prakash, Bhawna Paliwal, Mudit Dhawan, Neelabh Madan, Deepesh Hada, Vidit Jain, Sonu Mehta, Yashoteja Prabhu, Manish Gupta, Ramachandran Ramjee, Manik Varma
*International Conference on Learning Representations (**ICLR**), 2024*
Improves recommendations for rare products with few training examples; **deployed in Bing's advertising network.**

Reproducibility of AdaBelief Optimizer: Adapting Stepsizes by the Belief in Observed Gradients
Anirudh Buvanesh*, Madhur Panwar*
ReScience C, vol. 8, no. 2, 2022. ML Reproducibility Challenge 2021
Validates AdaBelief's fast, accurate, and stable training across diverse AI models; 🏆 **Spotlight**, NeurIPS 2022 Journal Track.

Modelling War Phenomena with an Emphasis on Operations Research
Anirudh Buvanesh, Sumanta Pasari
*IEEE International Conference on Computing, Communication and Networking Technologies (**ICCCNT**), 2021*
Uses mathematical models and simulations to study conflict and plan food supplies.

Preprints

On the Limits of Test-Time Compute for Self-Evaluation
Anirudh Buvanesh, Khaled Rouissi, Sujin Yun, Johan Obando-Ceron, Morgane M. Moss, Samuel Lavoie, Michael Noukhovitch, Laurent Charlin, Aaron Courville
Under review at ICLR 2027
Identifies a limit shared by existing test-time scaling algorithms and introduces Scorpio, which improves test-time scaling methods based on verification, tree search, and aggregation by up to 12 percentage points while reducing inference cost by up to 42%.

What Can You Do When You Have Zero Rewards During RL?
Jatin Prakash* and **Anirudh Buvanesh***
arXiv:2510.03971, 2025
Highlights how curricula can break the zero-reward barrier in reinforcement learning for LLMs; releases previously unavailable baseline implementations.

Patents

Retrieval of Novel Keywords for Search
Deepak Saini, Jian Jiao, Sachin Yadav, Bhawna Paliwal, **Anirudh Buvanesh**, Manik Varma
***US Patent 12,499,168**, granted December 2025. Assigned to Microsoft Technology Licensing*

Work Experience

Microsoft Research India *2022 – 2024*
Pre-doctoral Research Fellow, Extreme Classification Group
Developed algorithms for rare-class, zero-shot, and missing-label retrieval at the scale of hundreds of millions of products. All three methods were deployed in Microsoft products, including Bing.

Salesforce *2021 – 2022*
Software Engineer, Consumer Goods Cloud
Built and deployed microservices on Kubernetes to integrate an acquired trade-promotion product into the Salesforce platform.

Central Electronics Engineering Research Institute (CEERI) *Jan – Jul 2021*
Research Intern
Developed computer-vision methods for semantic segmentation of aerial farmland imagery.

Amazon *Aug – Dec 2020*
Software Engineering Intern, Supply Quality
Extended Amazon's advertising brand-safety engine with third-party signals and migrated Hadoop workloads to EMR.

Salesforce *May – Jul 2020*
Software Engineering Intern, Loyalty Cloud
Built a framework for complex business rules that scaled to one million rules; received a pre-placement offer.

Tata Institute of Fundamental Research *May – Jul 2018*
Research Intern, Gnowledge Lab
Developed a data-curation and analysis pipeline for the NROER education platform.

Awards and Scholarships

- **Bourse d'exemption, Université de Montréal** (2024–2027) — CAD 44,482 tuition waiver, awarded to attract top international doctoral students.
- **BITS Merit Scholarship** (2017–2021) — 40% tuition waiver, CAD 5,600, for placing in the top 2% of a cohort of 950 students.
- **Victor Menezes Scholarship** (2016–2020) — national award of CAD 9,000 for undergraduate study.
- **INSPIRE Scholarship**, Government of India (2016–2020) — CAD 5,000 for a top 1% rank in the Class XII examinations.
- **Joint Entrance Examination** (2016) — Ranked 780 among 150,000 candidates (top 0.4%) in India's national engineering exam.

Selected Projects

Université de Montréal and Mila *2024 – present*
Doctoral research. Advisors: Prof. Aaron Courville, Prof. Laurent Charlin

- Developed Scorpio, an algorithm that improves test-time scaling across verification, tree search, and aggregation methods.
- Demonstrated that curricula can overcome the zero-reward barrier when models cannot obtain rewards on difficult tasks.
- Current direction: improving credit assignment in reinforcement learning for large language models.

Microsoft Research India — Extreme Classification Group *2022 – 2025*
Research Intern and Collaborator. Advisors: Dr. Manik Varma, Dr. Manish Gupta, Dr. Amit Sharma

- Led two of three projects on retrieval systems that must choose among millions of categories, each addressing a different failure of the training signal: rare categories, unseen categories, and missing labels.
- All three shipped to Microsoft production systems serving millions of users daily.

Service, Teaching and Community

- **Reviewer** — ICLR 2025, KDD 2025, COLM 2026, ICLR 2026, ICLR 2027.
- **Mentor**, Fatima Fellowship (2026) — mentoring a student through an independent machine learning research project.
- **Open-source contribution** — released reference implementations of four reinforcement learning methods whose authors published no code (rl-reasoning/rl-baselines).
- **Reproducibility** — ML Reproducibility Challenge 2021; report selected as a spotlight at the NeurIPS 2022 Journal Track.
- **Reading group organizer**, Extreme Classification Group, Microsoft Research India (2022–2024) — weekly meetings on information retrieval and NLP, plus invited talks.
- **Teaching assistant**, *Logic in Computer Science*, BITS Pilani (2018).
- **Technical article** — *What Can You Do When You Have Zero Rewards During RL?*.